MIRZA AQIB BAIG

AI/ML Engineer | Python | Deep Learning | LLMs & RAG

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SUMMARY

AI/ML engineer (B.E. Computer Science, 9.0 CGPA) with hands-on production experience building and deploying ML systems end to end: a business-intelligence and profit-optimization platform on PostgreSQL and Power BI, a live RAG recruiter platform on FastAPI and FAISS, and a self-improving 207-class vehicle classifier with automated retraining. Strongest in PyTorch, LLM integration, vector search and Docker deployment.

TECHNICAL SKILLS

Languages: Python, SQL

ML & Deep Learning: PyTorch, Scikit-Learn, TensorFlow, Keras, EfficientNet-B3, transfer learning, fine-tuning, feature engineering, neural networks, demand forecasting

LLM & Generative AI: Retrieval-Augmented Generation (RAG), Sentence-Transformers, embeddings, semantic search, vector search (FAISS), HuggingFace Transformers, LLM APIs (OpenAI, Gemini, Groq), Ollama, Llama 3.2, prompt engineering, human-in-the-loop evaluation

Backend & APIs: FastAPI, Flask, REST API design, unit testing

Databases: PostgreSQL, MySQL, MS SQL Server — query optimization, stored procedures, data modeling

MLOps & Deployment: Docker, Railway, automated retraining pipelines, model monitoring, confidence thresholding, fallback guardrails

Tools & Workflow: Git, Jupyter Notebook, Pandas, NumPy, Power BI, Tableau, technical documentation

EXPERIENCE

Junior AI/ML Engineer | Digilitics | Hyderabad, India | Mar 2026 - Present

- Built an end-to-end business intelligence platform for a building-materials distribution client — PostgreSQL ingestion of sales, inventory and customer data, with Python and Pandas handling cleaning, transformation and feature generation — developed against public retail transaction datasets as a proxy while client data access was pending.
- Developed three ML modules on the shared warehouse: demand forecasting for inventory planning, customer scoring for account prioritization, and margin-level profit optimization surfacing loss-making product and customer segments.
- Modeled the Power BI reporting layer over the same schema, delivering profit, customer and inventory intelligence through a single owner-facing dashboard.

Data Scientist Intern | Full Stack Academy | Hyderabad, India | Oct 2025 - Feb 2026

- Built and evaluated supervised ML models in Python and Scikit-Learn, running data cleaning, feature engineering and hyperparameter tuning across structured datasets **[FILL: dataset size + metric lift, e.g. "raising F1 from 0.71 to 0.85 on a 5,000-record dataset"]**
- Containerized model inference services with Docker, producing reproducible environments and removing dependency conflicts between local and shared setups.
- Wrote and optimized SQL queries on MS SQL Server to extract and transform raw data into clean training sets feeding downstream model pipelines.
- Condensed current AI research and papers into written technical summaries for team knowledge sharing, folding structured review feedback into successive model versions.

PROJECTS

TalentIQ — AI-Powered Recruiter Intelligence Platform | Python, FastAPI, PostgreSQL, FAISS, Sentence-Transformers, RAG, Docker, Railway

Live: ai-talent-intelligence-platform-production.up.railway.app

- Built semantic resume-to-job matching by embedding candidates and roles with Sentence-Transformers and indexing them in a FAISS vector store, ranking matches through a weighted scoring algorithm across skills, experience and role fit.
- Implemented a RAG pipeline on Llama 3.2 via Ollama to generate written candidate-job fit analyses, with deterministic fallback logic keeping the endpoint responsive when the LLM is unavailable.
- Shipped the full backend on FastAPI and PostgreSQL with automated resume parsing, skill extraction and unit tests, deployed to Railway as a live public service.

Car Classifier — Self-Improving Vehicle Recognition Pipeline | Python, PyTorch, EfficientNet-B3, FastAPI, MySQL, LLM APIs

Repo: github.com/aqibbaigg/car_classifier

- Extended a PyTorch EfficientNet-B3 classifier from 196 to 207 vehicle classes through incremental fine-tuning on real-world images while preserving performance on the original classes **[FILL: top-1 accuracy on your test set]**
- Designed a confidence-thresholded fallback chain routing low-confidence predictions to Gemini, OpenAI and Groq APIs, backed by a MySQL review queue and human-in-the-loop validation before any label enters training **[FILL: share of low-confidence cases resolved]**
- Automated continuous retraining through auto_pipeline.py, triggering fine-tuning once 40+ human-approved samples accumulate — an end-to-end FastAPI MLOps loop that runs without manual intervention.

EDUCATION

B.E. Computer Science — Lords Institute of Engineering and Technology | Hyderabad | 2022 - 2026 | CGPA: 9.0/10

Certifications: Machine Learning — Internshala | Data Science | AI Fundamentals | Data Analytics **[FILL: issuer + year]**